Kaili Cao

Eric and Wendy Schmidt AI in Science Postdoctoral Fellow, Michigan Institute for Data and AI in Society (MIDAS), University of Michigan, Ann Arbor, Michigan, United States

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Positions

2026 – 2028: **Schmidt AI in Science Postdoctoral Fellow**, University of Michigan, Ann Arbor, United States

2024 – present: **Member**, Asteroseismology Using The Galactic Bulge Time Domain Survey (Roman Space Telescope Wide-Field Science Team), Columbus, United States

2023 – present: **Member**, Maximizing Cosmological Science with the Roman High Latitude Imaging Survey (Roman Space Telescope Project Infrastructure Team), Pasadena, United States

2023 – 2026: **Graduate Research Assistant**, Center for Cosmology and AstroParticle Physics (CCAPP), The Ohio State University, Columbus, United States

Education

2023 – 2026: **Doctor of Philosophy, Physics**, The Ohio State University, Columbus, United States
Dissertation: *Renovating Analysis Infrastructure for the Next Generation of Astronomical Surveys*

2021 – 2023: **Master of Science, Physics**, The Ohio State University, Columbus, United States

2020 – 2021: **31 credits, Computer Science and Technology**, Nanjing University, Nanjing, China

2016 – 2020: **Bachelor of Arts, French Language and Literature**, Nanjing University, Nanjing, China

Honors and Awards

2026 – 2028: **Eric and Wendy Schmidt AI in Science Postdoctoral Fellowship**, Michigan Institute for Data and AI in Society, University of Michigan.

2023 – 2026: Research support as a member of NASA's Roman HLIS Cosmology Project Infrastructure Team and the Roman asteroseismology Wide-Field Science Team (as a PhD student).

Research Interests

Cosmology: Weak Gravitational Lensing, Cosmic Shear Measurement, Parameter Inference, Fisher Forecasts

Stellar Physics: Stellar Models, Stellar Evolution Code, Interior Physics, Asteroseismology, Red Giant Branch

Methods: Numerical Simulations, Numerical Methods, Image Processing, Data Analysis, Machine Learning

Selected Publications

The full list (10 first-author and 14 coauthored papers, each with my contribution) is on kailicao.github.io/about/publications.html; ORCID: 0000-0002-1699-6944.

First-Author Publications

Oct 2026: **Kaili Cao**, David H. Weinberg, Vivian Miranda, et al., Fisher Forecasts for Cosmological Yields from 3×2pt Analysis of the Roman Space Telescope High Latitude Imaging Survey, *The Astrophysical Journal*, Volume 1010, Number 1, October 2026, Page 21, doi:10.3847/1538-4357/ae9bc0

Jul 2026: **Kaili Cao**, Efficient Optimal Image Reconstruction for the Nancy Grace Roman Space Telescope and Beyond: II. Implementation of EFFORTLESS, *arXiv e-prints*, arXiv:2607.09862 [astro-ph.IM], July 2026, doi:10.48550/arXiv.2607.09862 [GitHub]

Jul 2026: **Kaili Cao**, Efficient Optimal Image Reconstruction for the Nancy Grace Roman Space Telescope and Beyond: I. First Results with EFFORTLESS, *arXiv e-prints*, arXiv:2607.06674 [astro-ph.IM], July 2026, doi:10.48550/arXiv.2607.06674 [GitHub]

Mar 2026: **Kaili Cao** and Roman HLIS Cosmology PIT, Linear Image Regridding and Coaddition with Oversampled Point Spread Functions: Lessons from 1D, *The Astronomical Journal*, Volume 171, Number 3, March 2026, Page 140, doi:10.3847/1538-3881/ae356a [GitHub]

Feb 2026: **Kaili Cao**, Christopher M. Hirata, Katherine Laliotis, et al., Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. IV. Hyperparameter Optimization and Experimental Features, *The Astrophysical Journal*, Volume 998, Number 2, February 2026, Page 304, doi:10.3847/1538-4357/ae3a97 [GitHub]

Jan 2026: **Kaili Cao** and Marc H. Pinsonneault, Modeling APOKASC-3 Red Giants. I. The First Dredge-up and Red Giant Branch Bump, *The Astrophysical Journal*, Volume 997, Number 1, January 2026, Page 120, doi:10.3847/1538-4357/ae22d1 [GitHub, Zenodo]

Dec 2025: **Kaili Cao**, ContEvol Formalism: Numerical Methods Based on Hermite Spline Optimization, *Mathematics*, Volume 13, Issue 24, December 2025, Article Number 3981, doi:10.3390/math13243981 [GitHub]

Sep 2025: **Kaili Cao** and Marc H. Pinsonneault, Toward Accurate Asteroseismic Masses for Luminous Giants, *The Astrophysical Journal Letters*, Volume 990, Number 1, September 2025, Page L26, doi:10.3847/2041-8213/adfa12

Apr 2025: **Kaili Cao**, Christopher M. Hirata, Katherine Laliotis, et al., Simulating Image Coaddition with the Nancy Grace Roman Space Telescope. III. Software Improvements and New Linear Algebra Strategies, *The Astrophysical Journal Supplement Series*, Volume 277, Number 2, April 2025, Page 55, doi:10.3847/1538-4365/adb580 [GitHub]

May 2021: **Kaili Cao**, David J. Barnes, and Mark Vogelsberger, Studying galaxy cluster morphological metrics with MOCK-X, *Monthly Notices of the Royal Astronomical Society*, Volume 503, Issue 3, May 2021, Pages 3394–3413, doi:10.1093/mnras/stab605

Selected Presentations

2026

Jun 2026: **Renovating Analysis Infrastructure for the Next Generation of Astronomical Surveys (dissertation defense)**, *The Ohio State University*, Columbus, United States. [Slides]

Apr 2026: **Weak Gravitational Lensing Cosmology with the Roman Space Telescope High Latitude Imaging Survey (seminar)**, *Cosmology Group Seminar (University of Michigan)*, Ann Arbor, United States. [Slides]

Mar 2026: **Hot-wiring the Roman Sky: Resolving the Undersampling Crisis with Fast IMCOM for Real-Time Discovery (contributed talk)**, *RAPID Response: Hot-wiring the Next Generation of Time-Domain Science (Caltech/IPAC)*, Pasadena, United States. (Fast IMCOM is the earlier name of Effortless.) [Slides]

2025

Oct 2025: **Testing Stellar Theory with Joint Spectroscopic and Asteroseismic Catalogs (invited seminar talk)**, *AstroCoffee (University of Pittsburgh)*, Pittsburgh, United States. [Slides]

Oct 2025: **Toward Accurate Asteroseismic Masses for Luminous Giants (invited talk, online)**, *Galactic Genesis Working Group (SDSS-V Milky Way Mapper)*. [Slides]

Teaching Experience

2022 – 2023: **Instructor, Online Python Tutorial**
About 46 students in 4 cohorts. Gave lectures, designed exercises, and mentored personal projects.

2021 – 2023: **Teaching Associate (Grader), Physics 7501 & 7502: (Graduate) Quantum Mechanics**, The Ohio State University, Columbus, United States

Service

2026 – present: **Lead**, MapHub (the MIDAS arXiv Preprint Hub), a carpentry of the Schmidt AI in Science Fellowship, with Kyle Heyblom, Eva Albalghiti and Dimitrios Simatos.

2025: **Organizer**, PyIMCOM hack session for the Roman HLIS Cosmology Project Infrastructure Team.

Skills

Programming: Python (main), C/C++, Fortran, Shell/Perl, Wolfram Mathematica

Languages: Chinese (native), English (working proficiency), French (undergraduate major)